

Laboratory Assignment: DNS Server Deployment and DNS Delegation

Course: Unix-Based Internet Applications

Topic: DNS Server Configuration and Zone Delegation

Duration: 3-4 hours

Difficulty Level: Intermediate

Learning Objectives

By completing this laboratory assignment, students will be able to:

1. Install and configure BIND9 DNS server on a Linux system
2. Create and manage primary (master) DNS zones with various resource records
3. Configure secondary (slave) DNS servers for redundancy
4. Implement DNS zone delegation for subdomain management
5. Test and verify DNS configurations using diagnostic tools
6. Troubleshoot common DNS configuration issues

Prerequisites

- Basic understanding of DNS concepts (domains, zones, resource records)
- Familiarity with Linux command-line interface
- Root or sudo access to Linux systems (Ubuntu 20.04/22.04 or Debian 11 recommended)
- Basic text editor skills (nano, vim, or vi)
- Understanding of IP networking fundamentals

Lab Environment Setup

Required Resources

- **Three virtual machines** or physical systems:
 - **Server 1 (ns1):** Primary DNS server - IP: 192.168.100.10
 - **Server 2 (ns2):** Secondary DNS server - IP: 192.168.100.11
 - **Server 3 (ns3):** Delegated subdomain DNS server - IP: 192.168.100.12
- All systems should be able to communicate on the same network
- Minimum 1GB RAM and 10GB disk space per system

- Internet connectivity for package installation

Network Configuration

Domain Structure:

- Primary zone: example.lab
- Delegated subdomain: dept.example.lab

Part 1: Installing BIND9 DNS Server

Task 1.1: Install BIND9 Package

On **Server 1 (ns1)** and **Server 2 (ns2)**, execute the following commands:

```
# Update package repositories
sudo apt update

# Install BIND9 and related utilities
sudo apt install bind9 bind9utils bind9-doc dnsutils -y

# Verify BIND9 installation
named -v
```

Expected Output: BIND version information should be displayed (e.g., BIND 9.18.x).

Task 1.2: Understanding BIND9 Directory Structure

Familiarize yourself with the key BIND9 configuration files and directories:

- **/etc/bind/named.conf** - Main configuration file
- **/etc/bind/named.conf.options** - Server-wide options and ACLs
- **/etc/bind/named.conf.local** - Local zone definitions
- **/etc/bind/named.conf.default-zones** - Default zones (localhost, reverse)
- **/var/cache/bind/** - Zone files directory (default)
- **/var/log/named/** - Log files directory

Task 1.3: Configure DNS Server Options

Edit the options configuration file on **Server 1**:

```
sudo nano /etc/bind/named.conf.options
```

Add the following configuration:

```

acl "trusted" {
    192.168.100.0/24; # Allow queries from local network
    localhost;
    localnets;
};

options {
    directory "/var/cache/bind";

    # Allow queries from trusted networks only
    allow-query { trusted; };

    # Disable recursion for authoritative-only server
    recursion no;

    # Listen on IPv4 only
    listen-on { 192.168.100.10; 127.0.0.1; };

    # Disable zone transfers by default
    allow-transfer { none; };

    # DNSSEC validation
    dnssec-validation auto;

    # Forwarders (optional, for external queries)
    # forwarders {
    #     8.8.8.8;
    #     8.8.4.4;
    # };
};

```

Configuration Explanation:

- **ACL (Access Control List):** Defines trusted networks that can query the server ^[1] ^[2]
- **allow-query:** Restricts who can make DNS queries ^[3]
- **recursion:** Disabled for authoritative-only operation ^[41]
- **allow-transfer:** Controls zone transfer access ^[41]^[51]

Task 1.4: Verify Configuration Syntax

Always check syntax before restarting services:

```

# Check main configuration
sudo named-checkconf

# Check with verbose output
sudo named-checkconf -p

```

Note: If no errors are displayed, the configuration is syntactically correct ^[3]^[60].

Part 2: Creating Primary DNS Zone

Task 2.1: Define Forward Zone

Edit the local zones configuration on **Server 1**:

```
sudo nano /etc/bind/named.conf.local
```

Add the forward zone definition:

```
zone "example.lab" {  
    type master;  
    file "/etc/bind/zones/db.example.lab";  
    allow-transfer { 192.168.100.11; }; # Allow transfer to ns2  
    notify yes;  
};
```

Parameters Explained:

- **type master:** Designates this server as the primary authoritative source ^[42][47]
- **file:** Location of the zone data file ^[1]
- **allow-transfer:** Permits zone transfers to secondary server ^[41][51]
- **notify:** Sends notifications to secondary servers when zone changes ^[^]47]

Task 2.2: Define Reverse Zone

Add the reverse lookup zone in the same file:

```
zone "100.168.192.in-addr.arpa" {  
    type master;  
    file "/etc/bind/zones/db.192.168.100";  
    allow-transfer { 192.168.100.11; };  
    notify yes;  
};
```

Note: Reverse zones use the special `.in-addr.arpa` domain for IP-to-name resolution ^[1] ^[2].

Task 2.3: Create Zone Files Directory

```
sudo mkdir -p /etc/bind/zones
```

Task 2.4: Create Forward Zone File

Create and edit the forward zone file:

```
sudo nano /etc/bind/zones/db.example.lab
```

Add the following content:

```
$TTL      86400
@         IN      SOA      ns1.example.lab. admin.example.lab. (
                                2025111301      ; Serial (YYYYMMDDNN format)
                                3600              ; Refresh (1 hour)
                                1800              ; Retry (30 minutes)
                                604800           ; Expire (1 week)
                                86400 )          ; Negative Cache TTL (1 day)

; Name Servers
@         IN      NS       ns1.example.lab.
@         IN      NS       ns2.example.lab.

; A Records for Name Servers
ns1       IN      A        192.168.100.10
ns2       IN      A        192.168.100.11

; Mail Exchange Record
@         IN      MX       10 mail.example.lab.

; Host A Records
www       IN      A        192.168.100.20
mail      IN      A        192.168.100.25
ftp       IN      A        192.168.100.30
db        IN      A        192.168.100.35

; Alias (CNAME) Records
webmail   IN      CNAME    www.example.lab.
smtp      IN      CNAME    mail.example.lab.

; Delegation for subdomain
dept      IN      NS       ns3.example.lab.
ns3       IN      A        192.168.100.12
```

Zone File Components ^[1] ^[4] ^[2] ^[5]:

- **\$TTL:** Default time-to-live for records (in seconds)
- **SOA Record:** Start of Authority - defines zone properties
- **NS Records:** Nameserver records identifying authoritative servers
- **A Records:** Maps hostnames to IPv4 addresses
- **MX Record:** Mail exchange server with priority
- **CNAME Records:** Canonical name (alias) records
- **Delegation Records:** NS and glue A records for subdomain delegation ^[6] ^[7]

Task 2.5: Create Reverse Zone File

```
sudo nano /etc/bind/zones/db.192.168.100
```

Add the following content:

```
$TTL      86400
@         IN      SOA     ns1.example.lab. admin.example.lab. (
                                2025111301      ; Serial
                                3600              ; Refresh
                                1800              ; Retry
                                604800           ; Expire
                                86400 )          ; Negative Cache TTL

; Name Servers
@         IN      NS      ns1.example.lab.
@         IN      NS      ns2.example.lab.

; PTR Records (reverse mappings)
10        IN      PTR      ns1.example.lab.
11        IN      PTR      ns2.example.lab.
12        IN      PTR      ns3.example.lab.
20        IN      PTR      www.example.lab.
25        IN      PTR      mail.example.lab.
30        IN      PTR      ftp.example.lab.
35        IN      PTR      db.example.lab.
```

PTR Records: Enable reverse DNS lookups (IP to hostname) ^[1] ^[2].

Task 2.6: Validate Zone Files

Check zone file syntax using `named-checkzone`:

```
# Check forward zone
sudo named-checkzone example.lab /etc/bind/zones/db.example.lab

# Check reverse zone
sudo named-checkzone 100.168.192.in-addr.arpa /etc/bind/zones/db.192.168.100
```

Expected Output:

```
zone example.lab/IN: loaded serial 2025111301
OK
```

Task 2.7: Restart BIND9 Service

```
# Restart the service
sudo systemctl restart bind9

# Check service status
sudo systemctl status bind9

# Enable autostart on boot
sudo systemctl enable bind9
```

Part 3: Configuring Secondary (Slave) DNS Server

Task 3.1: Configure Secondary Server Options

On **Server 2 (ns2)**, edit the options file:

```
sudo nano /etc/bind/named.conf.options
```

Use similar configuration as the primary, but adjust the listen-on address:

```
options {
    directory "/var/cache/bind";
    listen-on { 192.168.100.11; 127.0.0.1; };
    allow-query { trusted; };
    recursion no;
    dnssec-validation auto;
};
```

Task 3.2: Define Secondary Zones

Edit the local zones configuration:

```
sudo nano /etc/bind/named.conf.local
```

Add secondary zone definitions:

```
zone "example.lab" {
    type slave;
    file "/var/cache/bind/db.example.lab";
    masters { 192.168.100.10; };
};

zone "100.168.192.in-addr.arpa" {
    type slave;
    file "/var/cache/bind/db.192.168.100";
};
```

```
masters { 192.168.100.10; };  
};
```

Secondary Zone Parameters ^[42]^[47]:

- **type slave:** Designates this as a secondary server
- **masters:** IP address of the primary server to transfer from
- **file:** Local cache file for transferred zone data

Task 3.3: Restart Secondary Server

```
sudo systemctl restart bind9  
sudo systemctl status bind9
```

Task 3.4: Verify Zone Transfer

Check that zones were transferred successfully:

```
# Check syslog for transfer messages  
sudo tail -f /var/log/syslog | grep named  
  
# Verify zone files were created  
ls -la /var/cache/bind/
```

Expected Output: You should see `db.example.lab` and `db.192.168.100` files created by zone transfer ^[41]^[43]^[45].

Task 3.5: Manual Zone Transfer Testing

From **Server 2**, manually request a zone transfer:

```
dig @192.168.100.10 example.lab AXFR
```

AXFR (Full Zone Transfer): Transfers the complete zone file from master to slave ^[43]^[45]^[48]^[51]^[56].

Part 4: Implementing DNS Delegation

DNS delegation allows you to assign responsibility for a subdomain to different nameservers. This is essential for distributed DNS management in large organizations ^[40]^[42].

Task 4.1: Understanding Delegation Concepts

Key Components of Delegation ^[6] ^[7]^[40]:

- **NS Records:** Point to the authoritative nameservers for the subdomain

- **Glue Records:** A records for the delegated nameservers (required when NS is within delegated zone)
- **Parent Zone:** Contains delegation records
- **Child Zone:** Managed independently on delegated nameservers

Task 4.2: Install BIND9 on Delegated Server

On **Server 3 (ns3)**, install BIND9:

```
sudo apt update
sudo apt install bind9 bind9utils bind9-doc dnsutils -y
```

Task 4.3: Configure Delegated Subdomain Zone

On **Server 3**, edit the options file:

```
sudo nano /etc/bind/named.conf.options
```

Configure similarly to other servers:

```
options {
    directory "/var/cache/bind";
    listen-on { 192.168.100.12; 127.0.0.1; };
    allow-query { any; };
    recursion no;
    dnssec-validation auto;
};
```

Task 4.4: Define Subdomain Zone

Edit the local zones file:

```
sudo nano /etc/bind/named.conf.local
```

Add the delegated subdomain zone:

```
zone "dept.example.lab" {
    type master;
    file "/etc/bind/zones/db.dept.example.lab";
    allow-transfer { none; };
};
```

Task 4.5: Create Subdomain Zone File

```
sudo mkdir -p /etc/bind/zones
sudo nano /etc/bind/zones/db.dept.example.lab
```

Add the subdomain zone content:

```
$TTL      86400
@         IN      SOA      ns3.example.lab. admin.dept.example.lab. (
                                2025111301      ; Serial
                                3600              ; Refresh
                                1800              ; Retry
                                604800           ; Expire
                                86400 )          ; Negative Cache TTL

; Name Server for subdomain
@         IN      NS       ns3.example.lab.

; Glue record (points back to parent zone)
ns3.example.lab.  IN  A      192.168.100.12

; Subdomain hosts
hr        IN      A        192.168.100.50
finance   IN      A        192.168.100.51
it        IN      A        192.168.100.52
admin     IN      A        192.168.100.53

; CNAME in subdomain
portal    IN      CNAME     hr.dept.example.lab.
```

Task 4.6: Validate and Restart

```
# Check zone file
sudo named-checkzone dept.example.lab /etc/bind/zones/db.dept.example.lab

# Check configuration
sudo named-checkconf

# Restart service
sudo systemctl restart bind9
```

Task 4.7: Verify Delegation in Parent Zone

Return to **Server 1 (ns1)** and verify the delegation records are present in `/etc/bind/zones/db.example.lab`:`

```
; Delegation for subdomain
dept     IN      NS       ns3.example.lab.
ns3      IN      A        192.168.100.12
```

These records were already added in Task 2.4. The NS record delegates authority, and the A record is the glue record [6] [7][^40].

Increment the serial number in the parent zone and restart:

```
sudo nano /etc/bind/zones/db.example.lab
# Change serial to 2025111302
sudo systemctl restart bind9
```

Part 5: Testing and Verification

Task 5.1: Basic DNS Query Testing

Use `dig` command to test DNS resolution from any system:

```
# Query the primary server
dig @192.168.100.10 www.example.lab

# Query the secondary server
dig @192.168.100.11 www.example.lab

# Query specific record types
dig @192.168.100.10 example.lab MX
dig @192.168.100.10 example.lab NS
dig @192.168.100.10 example.lab SOA
```

Expected Output: Should show the correct IP addresses and records [8] [^68].

Task 5.2: Test Subdomain Delegation

Query the delegated subdomain:

```
# Query subdomain from parent server (should be referred)
dig @192.168.100.10 hr.dept.example.lab

# Query subdomain from delegated server
dig @192.168.100.12 hr.dept.example.lab

# Check NS records for subdomain
dig @192.168.100.10 dept.example.lab NS
```

Verification: Queries to the parent should show referral to ns3, while queries to ns3 should return authoritative answers [40][42].

Task 5.3: Test Reverse DNS

```
# Reverse lookup
dig @192.168.100.10 -x 192.168.100.20

# Should return: www.example.lab
```

Task 5.4: Test Zone Transfer Security

Attempt zone transfer from an unauthorized host:

```
# This should FAIL if properly configured
dig @192.168.100.10 example.lab AXFR
```

Expected Result: Transfer should be refused, as zone transfers are restricted to ns2 only ^[41][51].

Task 5.5: Using nslookup for Testing

```
# Interactive mode
nslookup
> server 192.168.100.10
> www.example.lab
> set type=MX
> example.lab
> exit

# Non-interactive
nslookup www.example.lab 192.168.100.10
```

Task 5.6: DNS Trace Testing

Trace the full resolution path:

```
dig @192.168.100.10 hr.dept.example.lab +trace
```

This shows the complete delegation chain from root servers to your subdomain ^[68].

Task 5.7: Check DNS Server Logs

Monitor DNS activity:

```
# Real-time log monitoring
sudo tail -f /var/log/syslog | grep named

# Search for specific queries
sudo grep "query" /var/log/syslog | tail -20
```

Part 6: Advanced Configuration Tasks

Task 6.1: Configure DNS Logging

On **Server 1**, enhance logging capabilities. Edit the main configuration:

```
sudo nano /etc/bind/named.conf
```

Add a logging section:

```
logging {
    channel query_log {
        file "/var/log/named/query.log" versions 3 size 5m;
        severity info;
        print-time yes;
        print-severity yes;
        print-category yes;
    };

    channel transfer_log {
        file "/var/log/named/transfer.log" versions 3 size 5m;
        severity info;
        print-time yes;
    };

    category queries { query_log; };
    category xfer-in { transfer_log; };
    category xfer-out { transfer_log; };
};
```

Create log directory:

```
sudo mkdir -p /var/log/named
sudo chown bind:bind /var/log/named
sudo systemctl restart bind9
```

Logging Categories ^[1]:

- **queries:** Logs all DNS queries received
- **xfer-in/xfer-out:** Logs zone transfers

Task 6.2: Implement TSIG for Secure Zone Transfers

TSIG (Transaction Signatures) provides authentication for zone transfers using shared secret keys ^[59] ^{[62][^64]}.

On **Server 1**, generate a TSIG key:

```
# Generate HMAC-SHA256 key
sudo tsig-keygen -a hmac-sha256 ns1-ns2-key &gt; /etc/bind/tsig-keys.conf
```

View the generated key:

```
cat /etc/bind/tsig-keys.conf
```

Output example:

```
key "ns1-ns2-key" {
    algorithm hmac-sha256;
    secret "base64-encoded-secret-here==";
};
```

Include the key in named.conf:

```
sudo nano /etc/bind/named.conf
```

Add:

```
include "/etc/bind/tsig-keys.conf";
```

Modify the zone definition to use TSIG:

```
sudo nano /etc/bind/named.conf.local
```

Update the forward zone:

```
zone "example.lab" {
    type master;
    file "/etc/bind/zones/db.example.lab";
    allow-transfer { key ns1-ns2-key; };
};
```

On Server 2, copy the same TSIG key file and update configuration:

```
# Copy /etc/bind/tsig-keys.conf from Server 1 to Server 2
# (use scp or manually create with same secret)

sudo nano /etc/bind/named.conf
```

Add:

```
include "/etc/bind/tsig-keys.conf";
```

```
server 192.168.100.10 {
    keys { ns1-ns2-key; };
};
```

Restart both servers and test zone transfer with TSIG ^[59][64].

Task 6.3: Dynamic DNS Updates (Optional)

Configure a zone to accept dynamic updates (useful for DHCP integration):

```
sudo nano /etc/bind/named.conf.local
```

Add a dynamic zone:

```
zone "dynamic.example.lab" {
    type master;
    file "/var/cache/bind/db.dynamic.example.lab";
    allow-update { key ns1-ns2-key; };
};
```

Test dynamic update:

```
nsupdate -k /etc/bind/tsig-keys.conf
> server 192.168.100.10
> zone dynamic.example.lab
> update add test.dynamic.example.lab 300 A 192.168.100.99
> send
> quit
```

Part 7: Troubleshooting and Debugging

Common Issues and Solutions

Issue 1: BIND Service Fails to Start

Symptoms: Service status shows "failed" or "inactive"

Troubleshooting Steps:

```
# Check configuration syntax
sudo named-checkconf

# Check zone files
sudo named-checkzone example.lab /etc/bind/zones/db.example.lab

# Check system logs
sudo journalctl -xeu bind9
```

```
# Check for port conflicts
sudo netstat -tulpn | grep :53
```

Common Causes ^[3][^60]:

- Syntax errors in configuration files
- Invalid zone file format
- Port 53 already in use
- Permission issues on zone files

Issue 2: Zone Transfer Not Working

Symptoms: Secondary server doesn't receive zone updates

Troubleshooting ^[41][53]:

```
# On primary server, check if transfers are allowed
sudo grep allow-transfer /etc/bind/named.conf.local

# Check firewall rules
sudo ufw status
sudo ufw allow from 192.168.100.11 to any port 53

# Manually trigger transfer on secondary
sudo rndc retransfer example.lab

# Check transfer logs
sudo grep "transfer" /var/log/syslog
```

Issue 3: Delegation Not Resolving

Symptoms: Subdomain queries fail or return NXDOMAIN

Troubleshooting ^[40][42]:

```
# Verify NS records in parent zone
dig @192.168.100.10 dept.example.lab NS

# Verify glue records
dig @192.168.100.10 ns3.example.lab A

# Check delegation server is authoritative
dig @192.168.100.12 dept.example.lab SOA

# Trace resolution path
dig @192.168.100.10 hr.dept.example.lab +trace
```

Checklist:

- NS records properly configured in parent zone
- Glue A records present for delegated nameserver
- Delegated server is running and accessible
- Serial numbers are properly incremented

Issue 4: DNS Queries Timing Out

Troubleshooting [65][70][^73]:

```
# Check if server is listening
sudo ss -tulpn | grep :53

# Test connectivity
ping 192.168.100.10

# Check ACLs
sudo grep allow-query /etc/bind/named.conf.options

# Verify firewall settings
sudo iptables -L -n | grep 53
```

Useful Diagnostic Commands Summary

```
# Configuration validation
named-checkconf
named-checkconf -p # Print parsed config
named-checkzone <zone> <file>

# Service management
systemctl status bind9
systemctl restart bind9
rndc reload
rndc status
rndc retransfer <zone>

# Query tools
dig @<server> <domain> [type]
dig @<server> <domain> +trace
dig @<server> <zone> AXFR
nslookup <domain> <server>
host <domain> <server>

# Log monitoring
tail -f /var/log/syslog | grep named
journalctl -u bind9 -f
```

Lab Report Requirements

Students must submit a comprehensive lab report containing:

1. Executive Summary

- Brief overview of tasks completed
- Key learning outcomes

2. Configuration Documentation

- Complete copies of all configuration files:
 - named.conf.options
 - named.conf.local
 - Zone files (forward and reverse)
 - TSIG key configurations (if implemented)

3. Testing Results

- Screenshots or command output for:
 - DNS query tests (dig, nslookup)
 - Zone transfer verification
 - Delegation testing
 - Reverse DNS lookups
 - Error scenarios and troubleshooting

4. Network Diagram

- Diagram showing:
 - All DNS servers and their IP addresses
 - Zone hierarchy (parent and delegated zones)
 - Zone transfer relationships
 - Client query paths

5. Troubleshooting Section

- Document at least TWO problems encountered during the lab
- Describe the symptoms, diagnosis process, and resolution
- Include relevant log excerpts

6. Questions and Analysis

Answer the following questions in your report:

Question 1: Explain the difference between authoritative and recursive DNS servers. Which type did you configure in this lab? ^[9] ^[10] ^[11]

Question 2: What is the purpose of the Serial number in the SOA record? What happens if you forget to increment it when making zone changes? ^[1] ^[5] ^[38]

Question 3: Describe the zone transfer process. What are the differences between AXFR and IXFR? ^[43] ^[45] ^[48] ^[56]

Question 4: Why are glue records necessary in DNS delegation? What problems occur without them? ^[6] ^[7] ^[12]

Question 5: What security risks exist with improperly configured zone transfers? How did you mitigate these risks in your configuration? ^[41] ^[51]

Question 6: If the primary DNS server fails, can clients still resolve names in the zone? Explain why or why not. ^[42] ^[47]

Question 7: Describe three methods you used to troubleshoot DNS issues during this lab. ^[61] ^[63] ^[68] ^[70]

Question 8: What is the purpose of TSIG? In what scenarios would you implement it in a production environment? ^[59] ^[62] ^[64]

7. Reflection

- Personal insights gained from the lab
- Challenges faced and how you overcame them
- Real-world applications of DNS delegation

8. References

- List all documentation and resources consulted

Grading Rubric

Component	Points	Criteria
Primary DNS Configuration	20	Correct installation, zone files, and service operation
Secondary DNS Configuration	15	Proper slave setup and zone transfer verification
DNS Delegation	20	Successful subdomain delegation with NS and glue records
Testing and Verification	15	Comprehensive testing with multiple tools and methods
Advanced Configuration	10	TSIG implementation and logging configuration

Component	Points	Criteria
Troubleshooting	10	Documented problems and solutions
Lab Report	10	Complete documentation with all required sections
Total	100	

Additional Challenges (Bonus Points)

For advanced students seeking extra credit:

Challenge 1: IPv6 Support (+5 points)

- Configure AAAA records for IPv6 addresses
- Set up reverse DNS for IPv6 (ip6.arpa)
- Test IPv6 resolution

Challenge 2: DNSSEC Implementation (+10 points)

- Generate DNSSEC keys using `dnssec-keygen`
- Sign the zone with `dnssec-signzone`
- Configure DNSSEC validation
- Verify with `dig +dnssec [59][66][^71]`

Challenge 3: Split-Horizon DNS (+8 points)

- Configure different views for internal and external clients
- Use BIND views to provide different answers based on source IP
- Test from multiple network locations

Challenge 4: DNS Performance Monitoring (+5 points)

- Set up query rate limiting
- Configure response rate limiting (RRL)
- Monitor DNS query statistics with `rndc stats`

Safety and Best Practices

1. **Always validate configurations** before restarting services using `named-checkconf` and `named-checkzone [3][60][62]`
2. **Increment serial numbers** whenever zone files are modified [1][38][53]
3. **Use descriptive comments** in configuration files for maintainability
4. **Implement access controls** to prevent unauthorized queries and zone transfers [41][51]

5. **Regular backups** of zone files and configurations
6. **Monitor logs** for unusual activity and errors ^[1][61]
7. **Document changes** in a change log with dates and descriptions
8. **Test in isolated environments** before deploying to production

References and Further Reading

1. Red Hat Documentation - BIND DNS Server Configuration ^[1]
2. BIND 9 Administrator Reference Manual - Configurations and Zone Files ^[2]
3. DigitalOcean - Configure BIND as Private Network DNS Server ^[41]
4. CloudDNS - DNS Zones: Primary and Secondary ^[42]
5. RFC 1034 - Domain Names: Concepts and Facilities
6. RFC 1035 - Domain Names: Implementation and Specification
7. RFC 5936 - DNS Zone Transfer Protocol (AXFR) ^[48]
8. RFC 2845 - TSIG Protocol Specification ^[59][64]
9. ISC BIND 9 Documentation - <https://bind9.readthedocs.io/>
10. Computer Networking Notes - NS Record, Glue Record, and Delegation ^[7]

Appendix A: Quick Reference Commands

Service Management

```
sudo systemctl start bind9
sudo systemctl stop bind9
sudo systemctl restart bind9
sudo systemctl status bind9
sudo systemctl enable bind9
```

Configuration Validation

```
named-checkconf
named-checkconf -z
named-checkzone <zone-name> <zone-file>
```

RNDC Commands

```
rndc reload
rndc reload <zone>
rndc retransfer <zone>
```

```
rndc status
rndc flush
```

Query Tools

```
dig @<server> <name> [type]
dig +short <name>
dig +trace <name>
dig -x <ip-address>
dig AXFR <zone> @<server>
nslookup <name> <server>
host <name> <server>
```

Appendix B: Sample Zone File Template

```
$TTL      86400
$ORIGIN <domain>
@          IN      SOA      <primary-ns>. <admin-email>. (
                                <serial>          ; Serial
                                3600                ; Refresh
                                1800                ; Retry
                                604800               ; Expire
                                86400 )              ; Negative Cache TTL

; Name Servers
@          IN      NS       <ns1-name>.
@          IN      NS       <ns2-name>.

; Mail Servers
@          IN      MX       10 <mail-server>.

; Name Server A Records
<ns1-name> IN      A        <ns1-ip>
<ns2-name> IN      A        <ns2-ip>

; Host A Records
<hostname> IN      A        <ip-address>

; CNAME Records
<alias>    IN      CNAME    <canonical-name>.

; Subdomain Delegation
<subdomain> IN      NS      <delegated-ns>.
<delegated-ns> IN      A      <delegated-ip>
```

Lab Version: 1.0

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Instructor Contact: [Your contact information]

Good luck with your DNS server deployment and delegation configuration! Remember to document your work thoroughly and ask for assistance if you encounter difficulties.

[13] [14] [15] [16] [17] [18] [19] [20] [21] [22] [23] [24] [25] [26] [27] [28] [29] [30] [31] [32] [33] [34] [35] [36] [37]

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